



COMMONCOIN (COM)

The People's Cryptocurrency

A Technical Whitepaper detailing an independent, decentralized, and Script-based utility fork of Dogecoin Core.

Version: 1.0.0 (Official Technical Draft)

Date: June 20, 2026

Core Developers: CommonCoin Core Group

1. Abstract

CommonCoin (COM) is an open-source, peer-to-peer cryptocurrency designed to offer a decentralized utility network independent of legacy chains. Forked from Dogecoin Core stable version v1.14.9, CommonCoin implements specialized modifications to isolate its traffic, network topology, and genesis block variables. By rewriting connection port bindings, altering message start identifiers (magic bytes), and implementing custom base58 address prefixes, CommonCoin provides complete protocol independence. This whitepaper covers the consensus parameters, difficulty retargeting logic, block reward scheduling, and the deployment configuration of the CommonCoin blockchain.

2. Introduction & Motivation

The evolution of decentralized ledgers has demonstrated a strong market demand for accessible, transaction-friendly P2P digital currencies. CommonCoin was created to extend the proven codebase of Dogecoin Core into a community-centric ecosystem with modern developer telemetry, integrated mining pool systems, and absolute protocol isolation.

To achieve a successful launching state, CommonCoin establishes key parameters that distinguish it from the parent project:

- **Network Traffic Isolation:** Unique magic bytes prevent packet routing overlaps with Dogecoin or Litecoin, preventing peer-to-peer cross-talk.
- **Address Architecture:** Distinct base58 prefixes generate addresses starting with 'C', guaranteeing that wallets reject invalid transaction addresses.
- **Integrated Pool Infrastructure:** Native setup files and APIs are included in the repository for NOMP (Node Open Mining Pool), simplifying stratum pool launches.

3. Technical Specifications

The following parameters define the execution of the CommonCoin mainnet, testnet, and local regression testing (regtest) environments:

Parameter	CommonCoin (COM) Setting	Functional Role
Symbol / Ticker	COM	Token symbol utilized across wallets, exchanges, and interface.
Consensus Hashing	Script (Proof-of-Work)	Memory-hard mining algorithm to prevent ASIC centralization.
Block Target Time	1 Minute (60 Seconds)	Ensures fast confirmation times for consumer merchant apps.
Block Coinbase Reward	10,000 COM	Reward emitted to miners to secure consensus per solved block.
Mainnet P2P Port	33555	Default port used for peer-to-peer connections.
Mainnet RPC Port	33556	Secure command interface port (JSON-RPC daemon access).
Testnet P2P Port	44555	P2P network port for public beta testing and synchronization.
Testnet RPC Port	44556	RPC command execution port for test environments.
Address Prefix (P2PKH)	28 (Starts with 'C')	Base58 address prefix to prevent cross-network transactions.
Address Prefix (P2SH)	22 (Starts with 'A' or '9')	Allows secure multi-signature scripting addresses.
Message Start Bytes	0x434f4d4d (Mainnet)	Bytes prepended to packets to identify COM network packets.
Difficulty Adjustment	DigiShield (Per Block)	Calculates target difficulty dynamically on every solved block.

4. Genesis Block Parameters

To start the blockchain securely without reusing historical data, CommonCoin has generated independent genesis blocks using a custom coinbase string: *'CommonCoin - The People's Cryptocurrency'*. The calculated parameters are documented below:

Network	Nonce	Genesis Hash	Target Bits
Mainnet	3426683	0x95038c1832fde947069c67a846c6a2e8655351431bde617530e7ad5bfb4b0831e0ffff0	0831e0ffff0
Testnet	36396	0x0000008e62079c1bce6d7ec265b4c1adff633d909fdce3fbc3a528f162058002a1e0ffff0	02a1e0ffff0
Regtest	8	0x14ab6b1721c3a31598869ffc0ccdcc194caa928a23d726a8fda4a67061d8e612a07ffff	012a07ffff

5. Network Architecture & Sandbox Deployment

CommonCoin implements a structured local validator sandbox using virtual cloud infrastructure. This configuration ensures consensus stability, block propagation checks, and secure P2P syncing before WAN release. The deployment leverages two separate node processes executing on the same host, simulating separate nodes in a local loopback P2P link.

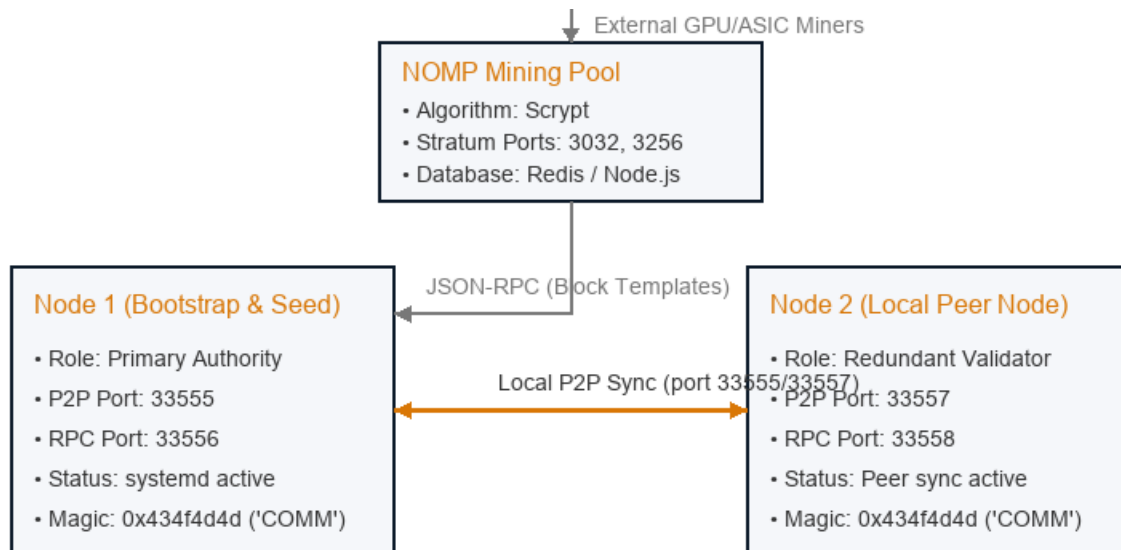


Figure 1: CommonCoin Local Sandbox P2P Sync & Mining Architecture.

As represented in the architecture diagram above, Node 1 is launched automatically as a systemd service daemon, listening on Mainnet P2P port 33555 and RPC port 33556. Node 2 is a separate process listening on P2P port 33557 and RPC port 33558, connecting to Node 1 via local loopback. Completed block templates are pulled via JSON-RPC, routed to the Node Open Mining Pool (NOMP) server, and solved blocks are propagated back to the network. This model demonstrates full-fledged block mining, transactions verification, and node sync.

6. DigiShield Difficulty Smoothing

A critical vulnerability for low-hashrate proof-of-work blockchains is difficulty manipulation by multipools. To mitigate this risk, CommonCoin integrates the **DigiShield difficulty adjustment algorithm**. DigiShield calculates block difficulty dynamically on every single block rather than using multi-hour windows. If a pool points massive hashpower to the network, the difficulty increases within one block. Once the pool leaves, the difficulty drops immediately, preventing the network from becoming stuck and ensuring smooth block propagation at the target 1-minute intervals.

7. Project Ecosystem Roadmap

CommonCoin's rollout is structured into distinct phases to guide it from a private sandbox to a decentralized public utility network:

Phase 1: Code Rebranding & Genesis Mining (Completed)

Codebase rebranded from Dogecoin Core. Mainnet, testnet, and regtest genesis blocks mined. First 111 blocks mined to secure consensus rewards on Node 1.

Phase 2: Private Sandbox & Local Validation (In Progress)

Local network established using VM instances. Automated deployments configured with Terraform and cloud-init templates. Wallet syncing verified.

Phase 3: Public Testnet & Client Wallets (Q3 2026)

Deploying seed servers to the public web. Releasing compiled Qt graphical wallets for Windows, macOS, and Linux users.

Phase 4: Stratum Mining Pools & Mainnet (Q4 2026)

Configuring NOMP pool stratum and public payouts. Launching the primary public bootstrap nodes and publishing block explorers.

8. Conclusion

CommonCoin introduces a clean, isolated, and accessible fork of Dogecoin Core. By modifying connection ports, address prefixes, and magic bytes, it establishes an independent consensus chain. With a dynamically adjusting difficulty algorithm (DigiShield), native mining pool settings, and robust cloud automation, CommonCoin represents a community-centered peer-to-peer transactional currency ready for public distribution.